Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2020	ENGI30520-WE01

Title: MSc Advanced Mechanical Engineering

Solid Mechanics 3

Time Allowed:	2 hours	
Additional Material provided:	NB - Formula sheets (5 pages) are included at the end of the paper.	
Materials Permitted:	None.	
Calculators Permitted:		Models Permitted: Those from the Casio fx-83 and fx-85 series.
Visiting Students may use dictionaries: YES		

Instructions to Candidates:	Answer ALL questions. All relevant workings must be shown. All questions carry equal marks.

Revision:	

Q1

(a) A thin high-strength steel sheet of width 70 mm is subjected to an oscillatory load characterized by a maximum stress of magnitude S and a minimum stress of 10 MPa as shown in Figure Q1a. A small local surface imperfection is detected at the upper side of the sheet that gives rise to a stress concentration effect that may be considered in the analysis using a fatigue strength reduction factor of 1.75. The S-N curve of the material is given in Figure Q1b for stress ratio $R = -1$. The fracture toughness of the high strength steel is 74 MPa$\sqrt{\text{m}}$ and its yield strength is 350 MPa. Taking into account mean stress effects, determine the maximum value of S in order to avoid fatigue failure. **[25%]**

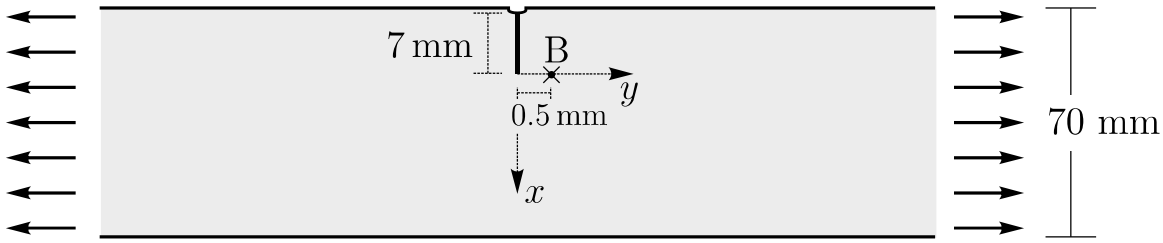


Figure Q1a: Steel thin sheet subjected to an oscillatory tensile stress. A local surface imperfection is detected at the upper side of the sheet. [Figure not to scale].

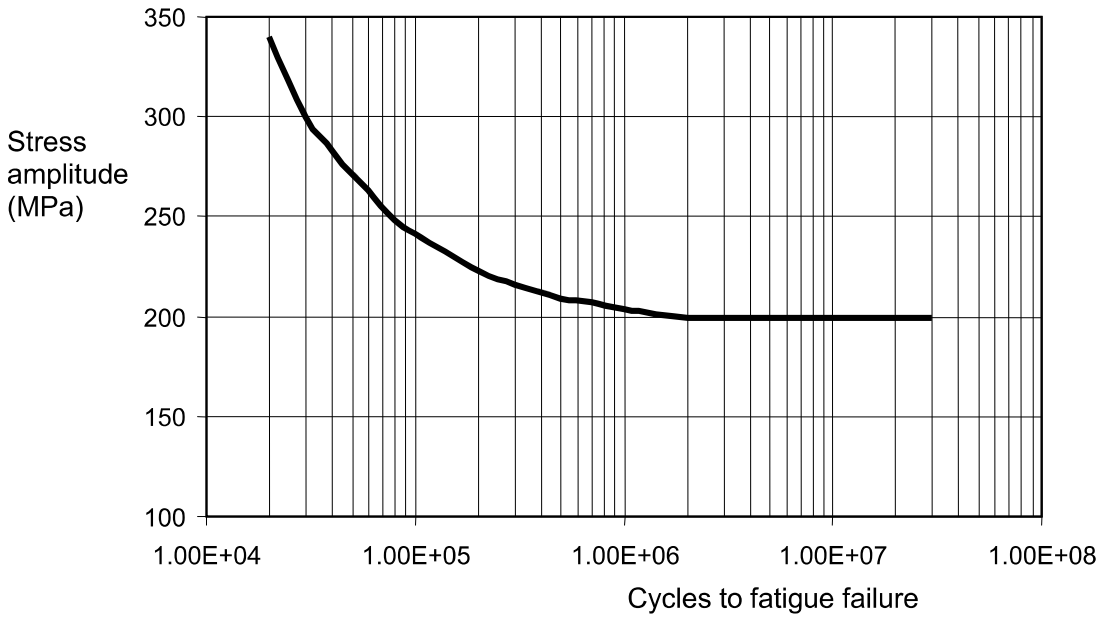


Figure Q1b: S-N curve of high-strength steel ($R = -1$).

(b) For safety reasons, the magnitude of the maximum stress is chosen to be $S = 150$ MPa. After some years of service, a 7 mm fatigue edge crack is detected that has developed from the stress concentration point as shown in Figure Q1a. The geometry factor Y for an edge crack in a finite plate is given in Figure Q1c. (i) Calculate all the stress components $(\sigma_{xx}, \sigma_{yy}, \sigma_{xy}, \sigma_{zz}, \sigma_{xz}, \sigma_{yz})$ at point B marked in Figure Q1a when the crack length is 7 mm and the tensile stress attains its maximum value. Assume that the (x, y)-axes are placed at the tip of the crack as shown in Figure Q1a. (ii) Determine the critical length of the crack for brittle fracture. **[35%]**

continued

(c) Using the Paris Law, and assuming the factor Y to take a constant value of the maximum it will achieve over the crack growth, determine the remaining fatigue life of the steel sheet for brittle fracture. The Paris Law coefficients are: $C = 9 \times 10^{-12}$ and $m = 2.4$ (assuming the use of MPa$\sqrt{\text{m}}$ for ΔK and m/cycle for da/dN). **[25%]**

(d) On closer examination of the cracked component, you find also evidence of corrosion on the steel sheet. Describe (i) how corrosion accelerates fatigue crack growth, and (ii) how fatigue accelerates corrosion damage. **[15%]**

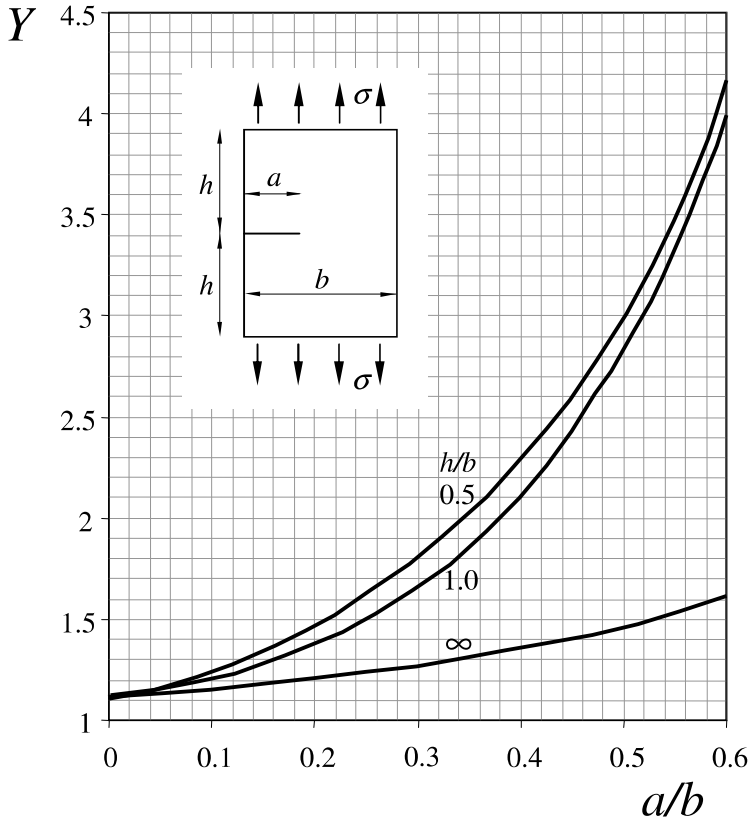


Figure Q1c: Y as function of the edge crack length in finite sheet under uniform tension.

Q2

Consider the 8-noded hexahedral finite element shown in Figure Q2 (a). The coordinates of each node are shown in brackets. The local coordinate system (ξ, η, ζ) and the node numbering convention are shown in Figure Q2 (b).

(a) Giving all of the steps in your derivation, show that the global derivatives of the shape functions are linked to the local derivatives via

$$\frac{\partial N_i}{\partial x} = \frac{2}{4}\frac{\partial N_i}{\partial \xi}, \qquad \frac{\partial N_i}{\partial y} = \frac{2}{3}\frac{\partial N_i}{\partial \eta} \qquad \text{and} \qquad \frac{\partial N_i}{\partial z} = \frac{2}{0.5}\frac{\partial N_i}{\partial \zeta},$$

where i is any one of the eight node numbers. Explain the physical meaning of the fractions in the above equations. **[35%]**

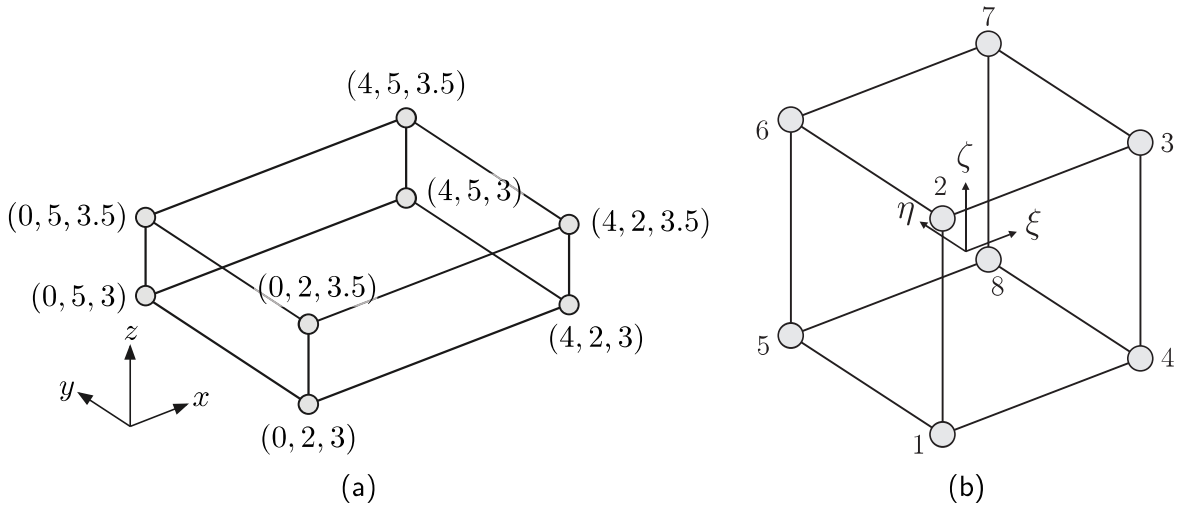


Figure Q2: 8-noded element: (a) global coordinates (in m) and (b) local node numbering

(b) Using equations on the formula sheet and those given in part Q2(a), determine the part of the strain-displacement matrix, $[B]$, associated with node 3 for a point in the element located at $(\xi, \eta, \zeta) = (0.1, -0.2, 0.8)$. **[30%]**

(c) The stiffness of a finite element can be determined using Gauss quadrature via

$$[k^e] = \sum_{i=1}^{n_{Gp}} [B]^T [D^e][B] \det([J]) w_i,$$

where n_{Gp} is the number of Gauss points. For the element shown in Figure Q2 (a) what is the determinant of the Jacobian matrix? Explain the physical meaning of this term. **[20%]**

(d) If the element in Figure Q2 (a) was halved in size in all directions what would happen to the values in the element stiffness matrix? Justify your answer using finite element theory/equations. **[15%]**

continued

Q3

Consider the finite element problem shown in Figure Q3 (a), where a rectangular specimen is subject to a horizontal displacement of 0.5m at one end whilst being fixed at the other and not allowing any vertical displacement on the top and bottom boundaries. The problem is meshed using two elements which have different Young's moduli.

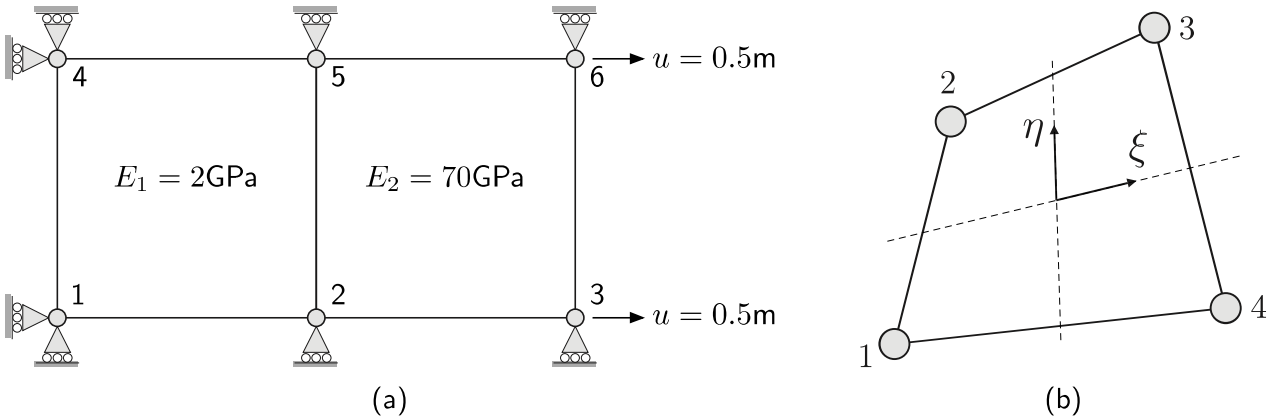


Figure Q3: (a) global mesh and (b) local node numbering

The element stiffness matrix, ordered according to the local convention shown in Figure Q3 (b), for the elements can be expressed as

$$[k^e] = E \begin{bmatrix} 0.509 & 0.174 & 0.046 & 0.035 & -0.255 & -0.174 & -0.301 & -0.035 \\ 0.174 & 0.509 & -0.035 & -0.301 & -0.174 & -0.255 & 0.035 & 0.046 \\ 0.046 & -0.035 & 0.509 & -0.174 & -0.301 & 0.035 & -0.255 & 0.174 \\ 0.035 & -0.301 & -0.174 & 0.509 & -0.035 & 0.046 & 0.174 & -0.255 \\ -0.255 & -0.174 & -0.301 & -0.035 & 0.509 & 0.174 & 0.046 & 0.035 \\ -0.174 & -0.255 & 0.035 & 0.046 & 0.174 & 0.509 & -0.035 & -0.301 \\ -0.301 & 0.035 & -0.255 & 0.174 & 0.046 & -0.035 & 0.509 & -0.174 \\ -0.035 & 0.046 & 0.174 & -0.255 & 0.035 & -0.301 & -0.174 & 0.509 \end{bmatrix}$$

where the matrix part of the above equation has units of m and E is the Young's modulus of the materials. In this case $E_1 = 2$GPa and $E_2 = 70$GPa.

(a) Show that the reduced stiffness matrix for this finite element mesh is **[40%]**

$$[K_{ff}] = \begin{bmatrix} 36.65 & 3.31 \\ 3.31 & 36.65 \end{bmatrix} \quad \text{GN/m.}$$

(b) Show that the forces in the x-direction at nodes 2 and 5 caused by the imposed displacement on nodes 3 and 6 are both 19.46GN. **[25%]**

(c) Determine the x displacement of nodes 5 and 2. Comment on the displacement of these nodes relative to nodes 3 and 6. **[25%]**

(d) How many Gauss points would you use to integrate the stiffness of each of the elements shown in Figure Q3 (a). Justify your answer. **[10%]**

continued

ENGI3411-WE01/ENGI30520-WE01 Solid Mechanics 3
Course: Fracture and Fatigue
Formula sheet

Stress and displacement distribution locally around the tip of a mode I crack oriented in the x-direction

$$\sigma_{xx} = \frac{K_I}{\sqrt{2\pi r}} \cos\frac{\theta}{2}\left[1 - \sin\frac{\theta}{2}\sin\frac{3\theta}{2}\right] + ...$$

$$\sigma_{yy} = \frac{K_I}{\sqrt{2\pi r}} \cos\frac{\theta}{2}\left[1 + \sin\frac{\theta}{2}\sin\frac{3\theta}{2}\right] + ...$$

$$\sigma_{xy} = \frac{K_I}{\sqrt{2\pi r}} \sin\frac{\theta}{2}\cos\frac{\theta}{2}\cos\frac{3\theta}{2} + ...$$

$$u_x = \frac{K_\mathrm{I}}{\mu}\sqrt{\frac{r}{2\pi}}\cos\frac{\theta}{2}\left[\frac{1}{2}(\kappa - 1) + \sin^2\frac{\theta}{2}\right] + \cdots$$

$$u_y = \frac{K_\mathrm{I}}{\mu}\sqrt{\frac{r}{2\pi}}\sin\frac{\theta}{2}\left[\frac{1}{2}(\kappa + 1) - \cos^2\frac{\theta}{2}\right] + \cdots$$

$\kappa = 3 - 4\nu$ (plane strain), or $(3-\nu)/(1+\nu)$ (plane stress)

Mode I stress intensity factor

$$K_I = Y\sigma\sqrt{\pi a}$$

Internal crack of length $2a$ in infinite plate: $Y = 1$
Edge crack of length a in relatively large plate: $Y = 1.12$

Plastic zone correction

$$\text{Equivalent crack length} = a + \frac{K_I^2}{2\pi\sigma_Y^2}$$

Failure diagrams for mean stress correction

Modified Goodman:

$$S_a = S_D\left[1 - \frac{S_{mean}}{S_u}\right]$$

Gerber parabola:

$$S_a = S_D\left[1 - \left(\frac{S_{mean}}{S_u}\right)^2\right]$$

Soderberg:

$$S_a = S_D\left[1 - \frac{S_{mean}}{S_{yield}}\right]$$

Strain-life method

Strain life equation:

$$\frac{\Delta\varepsilon}{2} = \frac{\sigma_f'}{E}(2N_f)^b + \varepsilon_f'(2N_f)^c$$

Mean stress correction (Smith Watson Topper):

$$\left(\sigma_{mean} + \frac{\Delta\sigma}{2}\right)\frac{\Delta\varepsilon}{2} = \frac{\sigma_f'^2}{E}(2N_f)^{2b} + \sigma_f'\varepsilon_f'(2N_f)^{b+c}$$

continued

Macro crack growth laws based on fracture mechanics approach

Paris Law:

$$\frac{da}{dN} = C(\Delta K)^m$$

Broek and Schijve:

$$\frac{da}{dN} = CK_{max}^2 \Delta K$$

Walker:

$$\frac{da}{dN} = CK_{max}^m (\Delta K)^n$$

Walker:

$$\frac{da}{dN} = C\overline{\Delta K}^n \qquad \text{where } \overline{\Delta K} = S_{max}(1-R)^m\sqrt{\pi a}$$

Forman:

$$\frac{da}{dN} = \frac{C(\Delta K)^m}{(1-R)(K_{Ic} - K_{max})}$$

continued

ENGI3411-WE01/ENGI30520-WE01 Solid Mechanics 3
Course: Finite-element method
Formula sheet

Three-dimensional stress-strain relationship

$$\begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{zz} \\ \sigma_{xy} \\ \sigma_{yz} \\ \sigma_{zx} \end{Bmatrix} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} (1-\nu) & \nu & \nu & 0 & 0 & 0 \\ \nu & (1-\nu) & \nu & 0 & 0 & 0 \\ \nu & \nu & (1-\nu) & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{(1-2\nu)}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{(1-2\nu)}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{(1-2\nu)}{2} \end{bmatrix} \begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{Bmatrix}$$

Two-dimensional stress-strain relationships

$$\begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{xy} \end{Bmatrix} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} (1-\nu) & \nu & 0 \\ \nu & (1-\nu) & 0 \\ 0 & 0 & \frac{(1-2\nu)}{2} \end{bmatrix} \begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{Bmatrix} \qquad \text{(plane strain)}$$

$$\begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{xy} \end{Bmatrix} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{(1-\nu)}{2} \end{bmatrix} \begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{Bmatrix} \qquad \text{(plane stress)}$$

Three-dimensional strain definition

$$\begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{Bmatrix} = [L]\{u\} = \begin{bmatrix} \frac{\partial}{\partial x} & 0 & 0 \\ 0 & \frac{\partial}{\partial y} & 0 \\ 0 & 0 & \frac{\partial}{\partial z} \\ \frac{\partial}{\partial y} & \frac{\partial}{\partial x} & 0 \\ 0 & \frac{\partial}{\partial z} & \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} & 0 & \frac{\partial}{\partial x} \end{bmatrix} \begin{Bmatrix} u \\ v \\ w \end{Bmatrix}$$

Interpolation using shape functions

$$u = \sum_{i=1}^{n_{en}} N_i u_i = \{N\}^T \{u\}$$

Three-dimensional strain-displacement definition

$$\begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{Bmatrix} = \underbrace{\begin{bmatrix} \frac{\partial N_1}{\partial x} & 0 & 0 & \dots & \frac{\partial N_{n_{en}}}{\partial x} & 0 & 0 \\ 0 & \frac{\partial N_1}{\partial y} & 0 & \dots & 0 & \frac{\partial N_{n_{en}}}{\partial y} & 0 \\ 0 & 0 & \frac{\partial N_1}{\partial z} & \dots & 0 & 0 & \frac{\partial N_{n_{en}}}{\partial z} \\ \frac{\partial N_1}{\partial y} & \frac{\partial N_1}{\partial x} & 0 & \dots & \frac{\partial N_{n_{en}}}{\partial y} & \frac{\partial N_{n_{en}}}{\partial x} & 0 \\ 0 & \frac{\partial N_1}{\partial z} & \frac{\partial N_1}{\partial y} & \dots & 0 & \frac{\partial N_{n_{en}}}{\partial z} & \frac{\partial N_{n_{en}}}{\partial y} \\ \frac{\partial N_1}{\partial z} & 0 & \frac{\partial N_1}{\partial x} & \dots & \frac{\partial N_{n_{en}}}{\partial z} & 0 & \frac{\partial N_{n_{en}}}{\partial x} \end{bmatrix}}_{[B]=[L][N]} \begin{Bmatrix} u_1 \\ v_1 \\ w_1 \\ \vdots \\ u_{n_{en}} \\ v_{n_{en}} \\ w_{n_{en}} \end{Bmatrix}$$

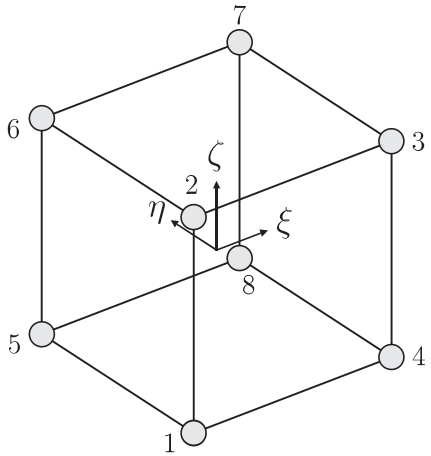
continued

Plane-stress/strain strain-displacement definition

$$\left\{\begin{array}{c}\varepsilon_{xx}\\ \varepsilon_{yy}\\ \gamma_{xy}\end{array}\right\} = \left[\begin{array}{cccc}\frac{\partial N_1}{\partial x} & 0 & \cdots & \frac{\partial N_{n_{en}}}{\partial x} & 0\\ 0 & \frac{\partial N_1}{\partial y} & \cdots & 0 & \frac{\partial N_{n_{en}}}{\partial y}\\ \frac{\partial N_1}{\partial y} & \frac{\partial N_1}{\partial x} & \cdots & \frac{\partial N_{n_{en}}}{\partial y} & \frac{\partial N_{n_{en}}}{\partial x}\end{array}\right] \left\{\begin{array}{c}u_1\\ v_1\\ \vdots\\ u_{n_{en}}\\ v_{n_{en}}\end{array}\right\}.$$

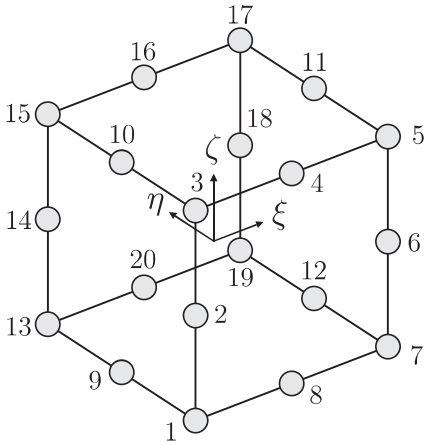
Shape functions for hexahedral elements

Tri-linear eight-noded hexahedral



$$\begin{aligned}
N_1 &= (1-\xi)(1-\eta)(1-\zeta)/8\\
N_2 &= (1-\xi)(1-\eta)(1+\zeta)/8\\
N_3 &= (1+\xi)(1-\eta)(1+\zeta)/8\\
N_4 &= (1+\xi)(1-\eta)(1-\zeta)/8\\
N_5 &= (1-\xi)(1+\eta)(1-\zeta)/8\\
N_6 &= (1-\xi)(1+\eta)(1+\zeta)/8\\
N_7 &= (1+\xi)(1+\eta)(1+\zeta)/8\\
N_8 &= (1+\xi)(1+\eta)(1-\zeta)/8
\end{aligned}$$

Tri-quadratic twenty-noded hexahedral



$$\begin{aligned}
N_1 &= (1-\xi)(1-\eta)(1-\zeta)(-\xi-\eta-\zeta-2)/8\\
N_2 &= (1-\xi)(1-\eta)(1-\zeta^2)/4\\
N_3 &= (1-\xi)(1-\eta)(1+\zeta)(-\xi-\eta+\zeta-2)/8\\
N_4 &= (1-\xi^2)(1-\eta)(1+\zeta)/4\\
N_5 &= (1+\xi)(1-\eta)(1+\zeta)(\xi-\eta+\zeta-2)/8\\
N_6 &= (1+\xi)(1-\eta)(1-\zeta^2)/4\\
N_7 &= (1+\xi)(1-\eta)(1-\zeta)(\xi-\eta-\zeta-2)/8\\
N_8 &= (1-\xi^2)(1-\eta)(1-\zeta)/4\\
N_9 &= (1-\xi)(1-\eta^2)(1-\zeta)/4\\
N_{10} &= (1-\xi)(1-\eta^2)(1+\zeta)/4\\
N_{11} &= (1+\xi)(1-\eta^2)(1+\zeta)/4\\
N_{12} &= (1+\xi)(1-\eta^2)(1-\zeta)/4\\
N_{13} &= (1-\xi)(1+\eta)(1-\zeta)(-\xi+\eta-\zeta-2)/8\\
N_{14} &= (1-\xi)(1+\eta)(1-\zeta^2)/4\\
N_{15} &= (1-\xi)(1+\eta)(1+\zeta)(-\xi+\eta+\zeta-2)/8\\
N_{16} &= (1-\xi^2)(1+\eta)(1+\zeta)/4\\
N_{17} &= (1+\xi)(1+\eta)(1+\zeta)(\xi+\eta+\zeta-2)/8\\
N_{18} &= (1+\xi)(1+\eta)(1-\zeta^2)/4\\
N_{19} &= (1+\xi)(1+\eta)(1-\zeta)(\xi+\eta-\zeta-2)/8\\
N_{20} &= (1-\xi^2)(1+\eta)(1-\zeta)/4
\end{aligned}$$

continued

Axi-symmetric strain-displacement definition

$$\begin{Bmatrix} \varepsilon_{rr} \\ \varepsilon_{zz} \\ \varepsilon_{\theta\theta} \\ \gamma_{rz} \end{Bmatrix} = \begin{bmatrix} \frac{\partial N_1}{\partial r} & 0 & \cdots & \frac{\partial N_{nen}}{\partial r} & 0 \\ 0 & \frac{\partial N_1}{\partial z} & \cdots & 0 & \frac{\partial N_{nen}}{\partial z} \\ \frac{N_1}{r_1} & 0 & \cdots & \frac{N_{nen}}{r_{nen}} & 0 \\ \frac{\partial N_1}{\partial z} & \frac{\partial N_1}{\partial r} & \cdots & \frac{\partial N_{nen}}{\partial z} & \frac{\partial N_{nen}}{\partial r} \end{bmatrix} \begin{Bmatrix} u_1 \\ w_1 \\ \vdots \\ u_{n_{en}} \\ w_{n_{en}} \end{Bmatrix}.$$

Global shape function derivatives

$$\begin{Bmatrix} \frac{\partial N_i}{\partial x} \\ \frac{\partial N_i}{\partial y} \\ \frac{\partial N_i}{\partial z} \end{Bmatrix} = \begin{bmatrix} \frac{\partial \xi}{\partial x} & \frac{\partial \eta}{\partial x} & \frac{\partial \zeta}{\partial x} \\ \frac{\partial \xi}{\partial y} & \frac{\partial \eta}{\partial y} & \frac{\partial \zeta}{\partial y} \\ \frac{\partial \xi}{\partial z} & \frac{\partial \eta}{\partial z} & \frac{\partial \zeta}{\partial z} \end{bmatrix} \begin{Bmatrix} \frac{\partial N_i}{\partial \xi} \\ \frac{\partial N_i}{\partial \eta} \\ \frac{\partial N_i}{\partial \zeta} \end{Bmatrix}.$$

The Jacobian matrix

$$[J] = \begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} & \frac{\partial z}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} & \frac{\partial z}{\partial \eta} \\ \frac{\partial x}{\partial \zeta} & \frac{\partial y}{\partial \zeta} & \frac{\partial z}{\partial \zeta} \end{bmatrix} = \begin{bmatrix} \frac{\partial N_1}{\partial \xi} & \frac{\partial N_2}{\partial \xi} & \cdots & \frac{\partial N_{nen}}{\partial \xi} \\ \frac{\partial N_1}{\partial \eta} & \frac{\partial N_2}{\partial \eta} & \cdots & \frac{\partial N_{nen}}{\partial \eta} \\ \frac{\partial N_1}{\partial \zeta} & \frac{\partial N_2}{\partial \zeta} & \cdots & \frac{\partial N_{nen}}{\partial \zeta} \end{bmatrix} \begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ \vdots & \vdots & \vdots \\ x_{n_{en}} & y_{n_{en}} & z_{n_{en}} \end{bmatrix}$$

Weak form of equilibrium

$$\int_V [B]^T [D^{\mathrm{e}}][B] dV \{d\} = \int_S [N]^T \{t\} dS + \int_V [N]^T \{f^b\} dV + \sum_{i=1}^{N} \{f\}_i$$

Element stiffness and force vector

$$[k^e] = \int_V [B]^T [D^{\mathrm{e}}][B] dV \qquad \text{and} \qquad \{f^e\} = \int_S [N]^T \{t\} dS + \int_V [N]^T \{f^b\} dV + \sum_{i=1}^{N} \{f\}_i.$$

Gauss-Legendre quadrature

$$\int_{-1}^{1} f(x) dx \approx \sum_{i=1}^{n_{Gp}} f(x_i) w_i$$

n_{Gp}	Abscissae, x_i	Weights, w_i
1	0.000000000000000	2.000000000000000
2	±0.577350269189626	1.000000000000000
3	0.000000000000000	0.888888888888889
	±0.774596669241483	0.555555555555556
4	±0.339981043584856	0.652145154862546
	±0.861136311594053	0.347854845137454

Numerical approximation of the element stiffness matrix

$$[k^e] \approx \sum_{i=1}^{n_{Gp}} [B]_i^T [D^{\mathrm{e}}][B]_i w_i \det([J]_i)$$

END